

Impact on the productivity of a manufacturing company using the proposed 5S methodology: a case study of furniture manufacturing

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Abstract

The 5S methodology is proposed for the sponge gluing area, identified as a bottleneck due to accumulation, waiting times, and disorganization. A diagnostic assessment was conducted (time study, process diagram, Ishikawa diagram, and Value Stream Mapping), and three alternatives were evaluated using discrete simulation in FlexSim. Proposal 1 resulted in higher productivity and reduced dwell time, improving the flow to upholstery.

Goals:

Objetivo General
To propose the 5S methodology in the sponge gluing area of a furniture manufacturing company, in order to improve productivity in the bottleneck.

Objetivo Específico

Clasificar el área aplicando la metodología 5S.

Diagnose the current situation of the foam joining area using Lean Manufacturing tools

Objetivo Específico

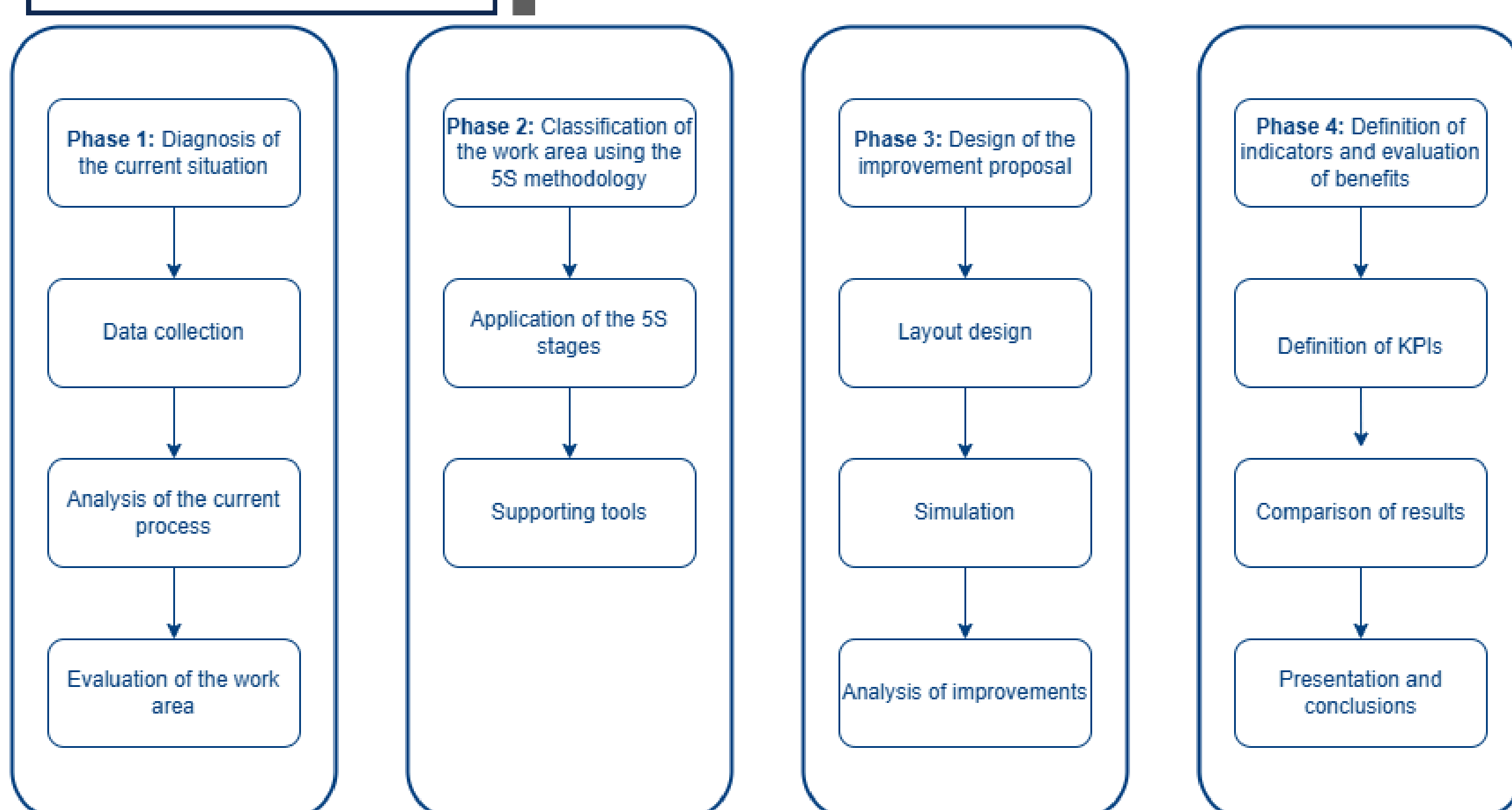
Evaluate the performance of the proposal using key indicators

Objetivo Específico

Design a proposal for improvement through the redistribution of the area

Objetivo Específico

Methodology



Curriculum Integration

- Lean Manufacturing and Six Sigma I in 5S Methodology
- CAD Industrial Design in Layout
- Process Engineering and Ergonomics in Time Studies for the Current Situation
- Occupational Health and Safety in Signage, Demarcation, and Visual Control

Applied Engineering

Engineering Standards:

- Executive Decree 255 for maintaining safe operating conditions
- Technical regulations on Occupational Safety and Health to establish order and cleanliness criteria
- NTE INEN-ISO 3864-1 for signage design
- NTE INEN-ISO 3864-2 for labeling and identification of storage areas
- NTE INEN 1467-1R for safety tags

Design constraints:

- Economic in the implementation budget
- Environmental in the handling of waste and adhesives
- Safety in the reduction of workplace hazards

keywords

5S Methodology

Performance indicators

bottleneck

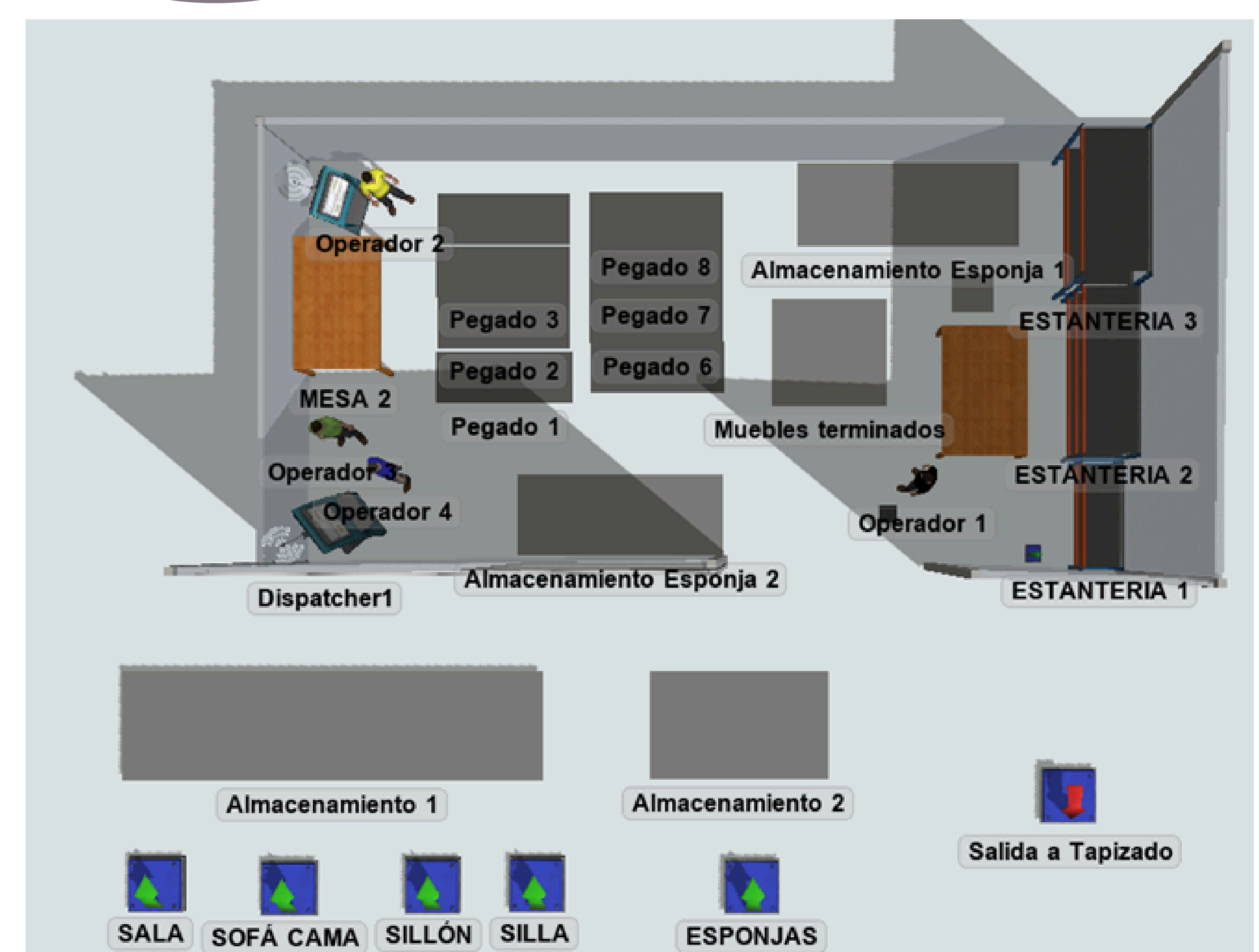
Discrete simulation

sponge gluing

Productivity

Results

Discrete simulation in FlexSim shows that Proposal 1 optimizes the overall system performance: it increases productivity from 1.7 to 2.7 units/hour and reduces the lead time of the flow, from 323.03 to 251.91 min for type 1 and from 563.52 to 358.04 min for type 2, reflecting less WIP accumulation and less waiting at the bottleneck of the sponge gluing area.



Conclusions

Conclusión

The evaluation using indicators (units completed, productivity, and dwell time) demonstrated that improvement depends not only on speed but also on reducing losses due to unnecessary movement and waiting times associated with disorganization and unnecessary movement. Therefore, the implementation of 5S with the area redistribution in Proposal 1 constitutes a solution for optimizing the flow; it reduces congestion in the sponge bonding process, improves process continuity to upholstery, and increases system performance under comparable operating conditions.

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References

- (Gupta, 2021)
- (Gonzalez et al., 2018)
- (Marmolejo et al., 2013)

